



Modelling Financial Markets

Chapter 13: Machine Learning in Finance



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Acronyms used in this material

- AI: Artificial Intelligence
- AUC: Area Under the (ROC) Curve
- CV: Cross-Validation
- DSR: Deflated Sharpe Ratio
- FFD: Fixed-width window Fractional Differentiation
- FN: False Negative
- GBM: Gradient Boosting Machine
- IS: In-Sample
- LASSO: Least Absolute Shrinkage and Selection Operator
- ML: Machine Learning
- MLP: MultiLayer Perceptron
- MSCA: Marie Skłodowska-Curie Actions
- OLS: Ordinary Least Squares
- OOB: Out-Of-Bag (error)
- OOS: Out-Of-Sample
- PAC: Probably Approximately Correct (learning)
- PBO: Probability of Backtest Overfitting
- PCR: Principal Component Regression
- PLS: Partial Least Squares
- PSR: Probabilistic Sharpe Ratio
- RF: Random Forest
- ROC: Receiver Operating Characteristic (curve)
- SR: Sharpe Ratio
- TN: True Negative
- TP: True Positive
- VIX: CBOE Volatility Index

Outline

Motivation

Machine Learning Foundations

Measuring Performance

Financial Data and Features

Labelling

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Case Study: S&P 500

Backtest Overfitting

Frontiers

Best Practices and Conclusions

References

Why Machine Learning in Finance?

The promise of machine learning (ML)

- ▣ Flexible, non-linear functional forms
- ▣ Hundreds of candidate predictors (the “factor zoo”, Chapter 3)
- ▣ Interactions discovered automatically
- ▣ Alternative data: text, images, order books, on-chain data

The problem

- ▣ Very low **signal-to-noise ratio**
- ▣ **Non-stationarity**: regimes, structural breaks
- ▣ Markets **adapt**: signals decay once exploited
- ▣ Few independent observations
- ▣ Massive risk of **overfitting** (fitting noise)

Key message of this chapter

- ▣ In finance, the hard part is not fitting a model — it is **not fooling yourself**: correct labels, correct validation, honest backtests

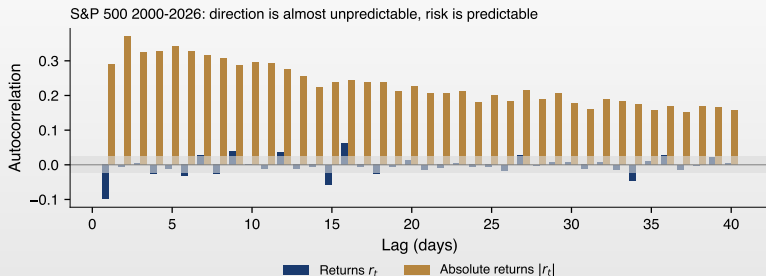
Finance Is Not Image Recognition

	Typical ML domain	Financial markets
Signal-to-noise	high (a cat is a cat)	extremely low
Data-generating process	stable	changes over time, reacts to users
Sample size	millions of i.i.d. examples	one history, serially dependent
Labels	given by humans	must be constructed
Cost of error	misclassification	real money, leverage, ruin
Experimentation	new data can be generated	cannot re-run history

Adaptive Markets Hypothesis: Lo (2004)

- Predictability is not a constant: it appears, is arbitrated away, and re-appears under new conditions
- A model that worked yesterday is a **hypothesis**, not a law (market efficiency: Chapter 2)

What Is Predictable? Direction vs. Risk



- Autocorrelation of lag k : $\rho_k = \text{Corr}(x_t, x_{t-k})$; the grey band is the 95% no-correlation band $\pm 1.96/\sqrt{T}$
- Autocorrelations of returns r_t are tiny (inside the band); those of $|r_t|$ are large and persistent
- ML has the best chance where there is structure: **volatility, risk, liquidity, cross-sectional ranking**

Evidence from Empirical Asset Pricing

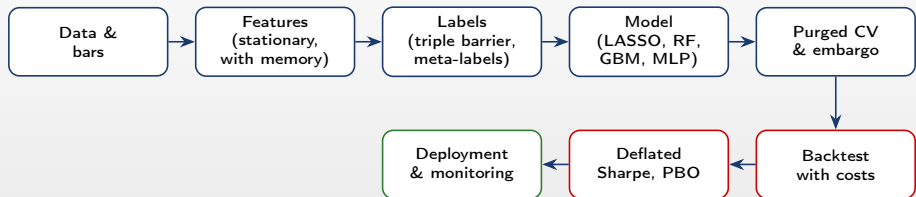
Gu, Kelly & Xiu (2020), *Review of Financial Studies*

- Predict **monthly returns of individual American stocks**; compared: linear models, penalised regressions, principal component regression (PCR), partial least squares (PLS), random forests, boosted trees, neural networks
- Out-of-sample $R_{OOS}^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t r_t^2$ is small (well below 1% per month)
- Trees and neural nets beat linear models; gains come from **non-linear interactions**; dominant predictors: momentum, liquidity, volatility
- A small R_{OOS}^2 can still rank thousands of stocks well enough for large portfolio gains

Caveat

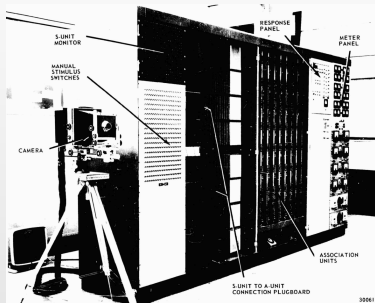
- For a **single time series** (one index) the effective sample is far smaller and the evidence much weaker — as the S&P 500 case study shows

The Financial ML Workflow



- Sections 4–6 follow the top row, Sections 7–8 the bottom row
- **Look-ahead bias**: using information not available at decision time; **selection bias**: reporting the best of many trials
- **López de Prado (2018)**: most ML failures in finance are due to the process, not to the algorithm

From the Perceptron to Deep Blue



Mark I Perceptron, operator's manual (1960)

Figure: J.C. Hay, A.E. Murray (1960); public domain; Wikimedia Commons



IBM Deep Blue, Computer History Museum

Photo: Anton Chiang (2011); CC BY 2.0; Wikimedia Commons

- **Perceptron**: a learning rule for a single artificial neuron [Rosenblatt \(1958\)](#); the Mark I was hardware built to recognise simple images
- 1997: IBM's Deep Blue beats world chess champion Garry Kasparov [Campbell, Hoane & Hsu \(2002\)](#)
- Deep Blue searched with hand-built rules; modern ML (this chapter) *learns* the rules from data

Supervised Learning: Setup

Definition 1 (Supervised learning)

- ▣ Data $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$: features (predictors) $\mathbf{x}_i \in \mathbb{R}^p$ and target (label) y_i
- ▣ Find the function $\hat{f}$ that predicts well on **new** data by minimising

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n L(y_i, f(\mathbf{x}_i)) + \lambda \Omega(f)$$

- ▣ L : loss function; $\Omega(f)$: penalty on complexity; $\lambda \geq 0$: tuning parameter
- ▣ The first term measures fit, the second prevents the model from memorising noise
- ▣ λ is not estimated from the training loss: it is chosen by cross-validation

Regression and Classification

Regression

- Target y continuous: next-period return, volatility
- Squared-error loss: $L = (y - f)^2$
- Prediction: conditional mean $\mathbb{E}[y \mid x]$

Classification

- Target $y \in \{0, 1\}$ or $\{-1, 0, 1\}$: direction, event
- Log-loss: $L = -y \log p - (1 - y) \log(1 - p)$,
 $p = \hat{P}(y = 1 \mid x)$
- Prediction: class probability p , class
 $= 1\{p > 0.5\}$

- This chapter uses classification: “will the price be higher in 5 days?”
- Logistic regression (logit) models $p = 1/(1 + e^{-x^\top \beta})$ — the linear benchmark

In-Sample, Out-of-Sample, Cross-Validation

- ▣ **In-sample (IS)** error: computed on the data used for fitting — always optimistic
- ▣ **Out-of-sample (OOS)** error: computed on data not used for fitting — what matters
- ▣ **Cross-validation (CV)**, K -fold: split the data into K blocks; fit on $K - 1$, evaluate on the remaining one; repeat K times and average
- ▣ CV is used to choose tuning parameters (λ , tree depth) and to compare models
- ▣ A final **test set**, never touched during model selection, gives the honest estimate

Finance-specific warning

- ▣ Standard K -fold CV assumes i.i.d. observations — Section 6 shows how this breaks for time series

Bias–Variance Trade-off

Proposition 1 (Decomposition of expected prediction error)

□ For $y = f(x) + \varepsilon$, $\mathbb{E}[\varepsilon] = 0$, $\text{Var}(\varepsilon) = \sigma^2$:

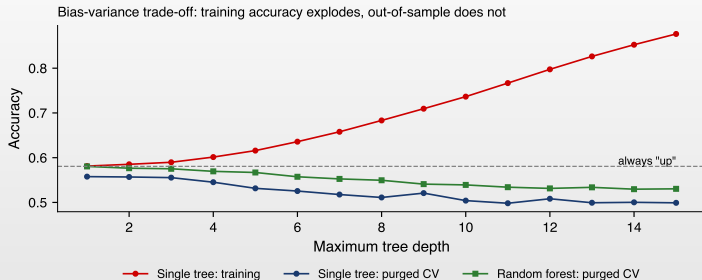
$$\mathbb{E}[(y - \hat{f}(x))^2] = \underbrace{(\mathbb{E}[\hat{f}(x)] - f(x))^2}_{\text{bias}^2} + \underbrace{\text{Var}(\hat{f}(x))}_{\text{variance}} + \underbrace{\sigma^2}_{\text{irreducible}}$$

- **Bias**: error from a model that is too simple (misses structure)
- **Variance**: error from a model that reacts to the particular sample (fits noise)
- **Irreducible** noise σ^2 : cannot be removed by any model

Bias–Variance in Finance

- ▣ In finance the noise σ^2 dominates the total error: most of the variation of returns is unpredictable
- ▣ Complex models reduce bias but inflate variance $\Rightarrow$ they **fit the noise**
- ▣ With little signal, a small reduction of bias is rarely worth a large increase in variance
- ▣ Tools that control variance:
 - ▶ regularisation (penalties $\lambda \Omega(f)$)
 - ▶ ensembles (averaging many models)
 - ▶ proper, leakage-free validation

Bias–Variance in Practice: Tree Depth



- S&P 500, 5-day direction; accuracy = share of correctly classified days
- Training accuracy rises to $\approx 87\%$ with depth; purged-CV accuracy **falls** towards 50%
- Random forests degrade more slowly, but never beat the “always up” baseline (dashed line)

Penalised Regression: Ridge, LASSO, Elastic Net

Definition 2 (Penalised least squares)

- ▣ Objective: $\hat{\beta} = \arg \min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \left[\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2 \right]$
 - ▣ Norms: $\|\beta\|_1 = \sum_j |\beta_j|$, $\|\beta\|_2^2 = \sum_j \beta_j^2$
 - ▣ $\alpha = 0$: **Ridge**, Hoerl & Kennard (1970) — shrinks, keeps all predictors
 - ▣ $\alpha = 1$: **LASSO**, Tibshirani (1996) — shrinks and sets coefficients exactly to zero
 - ▣ $0 < \alpha < 1$: **Elastic Net**, Zou & Hastie (2005) — handles groups of correlated predictors
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- ▣ For classification the squared error is replaced by the log-loss (penalised logit)

Why Shrinkage Helps, and How to Tune It

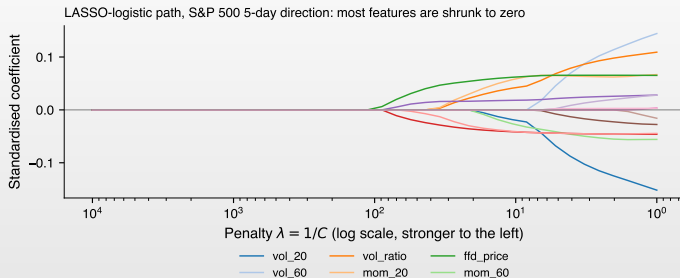
Why it helps in finance

- ▣ Many correlated predictors (momentum at several horizons, several volatility measures)
- ▣ Ordinary least squares (OLS) is then unstable: large, offsetting coefficients
- ▣ Shrinkage trades a little bias for much lower variance

Tuning

- ▣ λ chosen by (purged!) cross-validation
- ▣ Standardise features first (mean 0, standard deviation 1): the penalty is not scale-invariant
- ▣ Scaling parameters estimated on the training folds only

LASSO Regularisation Path

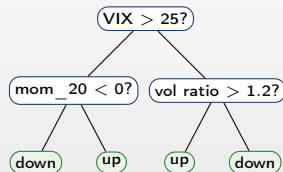


- L_1 -penalised logistic regression for the 5-day direction of the S&P 500, 14 standardised features
- Strong penalty: every coefficient is zero; the first feature to enter is the FFD price
- Weak penalty: the volatility features (vol_20, vol_60, vol_ratio) get the largest coefficients
- All coefficients are small in absolute value: weak signal, consistent with near-efficient markets

Decision Trees

Definition 3 (Classification tree)

- Recursive binary partition of the feature space into regions $R_1, \dots, R_M$
- Prediction = majority class (or mean) in each region:
$$\hat{f}(x) = \sum_{m=1}^M c_m 1\{x \in R_m\}$$
- Splits chosen greedily to minimise impurity, e.g. Gini
$$G = \sum_k p_k(1 - p_k), p_k = \text{class share in the node}$$



- An illustrative tree (not estimated)

- + Non-linearities and interactions, invariant to monotone transformations, easy to explain
- High variance: a small change in the data gives a very different tree $\Rightarrow$ average many trees (forest) or add small trees sequentially (boosting)

Random Forests

Random forest (RF): Breiman (2001)

- ▣ **Bagging** (bootstrap aggregating): grow B trees, each on a bootstrap sample of the data
- ▣ At each split only a random subset of $m \approx \sqrt{p}$ features is considered $\Rightarrow$ decorrelated trees
- ▣ Forest prediction = average of the trees: $\hat{f}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$
- ▣ Averaging reduces variance; out-of-bag (OOB) observations give a built-in error estimate

Finance caveat

- ▣ Bootstrap of **overlapping** labels is not i.i.d. $\Rightarrow$ the OOB error is optimistic



Worked Example: How a Forest Averages Its Trees

- ▣ Random forest of the case study (400 trees, minimum leaf size 100), trained on the first four purged folds
- ▣ Prediction for the last observation (11 September 2026): $P(\text{up in 5 days})$ from each tree
 - ▶ individual trees range from 0.31 to 0.88 (standard deviation 0.11)
 - ▶ 81.5% of the trees vote “up” ($p > 0.5$)
 - ▶ forest probability = average of the 400 trees = 0.608
- ▣ Single trees disagree strongly; the average is far more stable
- ▣ Stability is not accuracy: the forest still does not beat “always up” (Section 7)

Gradient Boosting

Gradient boosting machine (GBM): Friedman (2001)

- ▣ Additive model built sequentially:

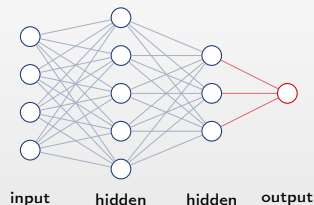
$$F_m(x) = F_{m-1}(x) + \eta h_m(x), \quad m = 1, \dots, M$$

- ▣ h_m : a small tree fitted to the negative gradient of the loss (the current errors)
 - ▣ $\eta \in (0, 1]$: learning rate; with the depth and the number of trees M it controls complexity
 - ▣ XGBoost, LightGBM, CatBoost: fast implementations, state of the art on tabular data
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- ▣ RF reduces variance by averaging; GBM reduces bias by correcting errors step by step

Neural Networks

Definition 4 (Multilayer perceptron (MLP), L layers)

- $h^{(0)} = x, \quad h^{(\ell)} = g(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)})$
 - Output: $\hat{y} = \sigma(w^\top h^{(L)} + b)$
 - g : rectified linear unit (ReLU), $g(z) = \max(0, z)$;
 $\sigma(z) = 1/(1 + e^{-z})$
 - Weights learned by stochastic gradient descent
-
- Regularisation: weight decay (L_2 penalty), dropout, **early stopping** (stop when the validation loss stops improving)
 - Shallow networks (1–3 layers) work best on tabular return data (Gu, Kelly & Xiu (2020)); depth pays off on text and order books (Chapters 14–15)



Recap: Machine Learning Foundations

- ▣ Supervised learning = loss + complexity penalty, tuned by cross-validation
- ▣ Out-of-sample error is the only relevant error
- ▣ Finance is noise-dominated: variance control matters more than flexibility
- ▣ Toolbox: penalised logit (Ridge / LASSO / Elastic Net), trees, random forests, gradient boosting, MLP
- ▣ Next: how to **measure** whether a classifier is any good

The Confusion Matrix and Accuracy

	Predicted up	Predicted down
Actual up	TP	FN
Actual down	FP	TN

- TP: true positives; FN: false negatives
- FP: false positives; TN: true negatives
- $n = TP + FN + FP + TN$

Definition 5 (Accuracy)

- Share of correct predictions: $\text{Accuracy} = \frac{TP + TN}{n}$
- With unbalanced classes accuracy is misleading: always predicting the majority class already scores its share

Precision, Recall and F1

Definition 6 (Precision, recall, F1 score)

- ▣ **Precision:** share of predicted “up” that were really up, $\text{Prec} = \frac{TP}{TP + FP}$
 - ▣ **Recall** (hit rate): share of actual “up” that were detected, $\text{Rec} = \frac{TP}{TP + FN}$
 - ▣ **F1 score:** harmonic mean, $F_1 = \frac{2 \text{Prec} \cdot \text{Rec}}{\text{Prec} + \text{Rec}}$
-
- ▣ In trading, precision = share of the trades you take that pay off
 - ▣ Recall = share of the good opportunities you actually take
 - ▣ Meta-labelling (Section 5) trades recall for precision

Worked Example: Confusion Matrix of the Logit Model

	Pred. up	Pred. down
Actual up	3,572	68
Actual down	2,591	38

- L2-penalised logit, S&P 500 case study, out-of-fold predictions of purged 5-fold CV, $n = 6,269$

- Accuracy = $(3,572 + 38)/6,269 = 57.6\%$
- Precision = $3,572/(3,572 + 2,591) = 0.580$
- Recall = $3,572/(3,572 + 68) = 0.981$
- $F_1 = 0.729$
- The model predicts “up” on 98.3% of days: it has essentially learned “always up”

ROC Curve and AUC

Definition 7 (ROC curve and AUC)

- For a threshold c , classify “up” if $\hat{p} > c$; true positive rate $TPR(c) = TP / (TP + FN)$, false positive rate $FPR(c) = FP / (FP + TN)$
- The **ROC curve** plots $TPR(c)$ against $FPR(c)$ for all $c \in [0, 1]$
- **AUC** = area under the ROC curve = $P(\hat{p}_{\text{pos}} > \hat{p}_{\text{neg}})$: probability that a random “up” day gets a higher score than a random “down” day
- $AUC = 0.5$: no ranking ability (coin flip); $AUC = 1$: perfect ranking
- AUC does not depend on the threshold or on class balance — report it with accuracy

Always Compare with a Naive Baseline

- ▣ “**Always up**” baseline: predict the majority class every day
- ▣ Its accuracy equals the share of up-labels: $\bar{y} = \frac{1}{n} \sum_i y_i$
- ▣ S&P 500, 5-day direction, 2001–2026: $\bar{y} = 58.1\%$ — a “57%-accurate” model is *worse* than doing nothing
- ▣ Other baselines: yesterday’s direction, a moving-average rule, buy-and-hold for trading
- ▣ A model has value only if it beats the baseline **out of sample**, after costs

From Predictions to a Trading Strategy

Definition 8 (Positions, turnover, costs, exposure)

- Position $w_t \in \{0, 1\}$: long the index (1) or in cash (0), decided with information up to t
 - Strategy return: $r_{t+1}^s = w_t r_{t+1} - c |w_t - w_{t-1}|$
 - Turnover $|w_t - w_{t-1}|$: fraction of the portfolio traded; cost c in basis points (5 bp = 0.05% per unit traded)
 - Exposure: average share of time invested, $\bar{w} = \frac{1}{T} \sum_t w_t$
-
- Buy-and-hold: $w_t = 1$ for all t (exposure 100%, no turnover)

The Sharpe Ratio

Definition 9 (Sharpe ratio (SR))

- Per-period SR of returns $r_1, \dots, r_T$ (excess over the risk-free rate, here set to 0):

$$\widehat{SR} = \frac{\bar{r}}{\hat{\sigma}}, \quad \bar{r} = \frac{1}{T} \sum_t r_t, \quad \hat{\sigma}^2 = \frac{1}{T-1} \sum_t (r_t - \bar{r})^2$$

- **Annualised SR:** $\widehat{SR}_{ann} = \sqrt{q} \widehat{SR}$, with $q = 252$ trading days (stocks) or $q = 365$ days (crypto)
- Mean return per unit of risk; scale-free, so strategies with different leverage are comparable
- The $\sqrt{q}$ rule assumes i.i.d. returns: an approximation

Recap: Measuring Performance

- ▣ Accuracy alone is misleading with unbalanced classes: always report the baseline
- ▣ Precision and recall separate “how often right when acting” from “how many opportunities taken”
- ▣ AUC measures ranking ability; $0.5 = \text{no skill}$
- ▣ Trading evaluation: Sharpe ratio after transaction costs, compared with buy-and-hold
- ▣ Next: how to build features and labels that do not leak information

Sampling Financial Data: Bars

Time and tick bars

- ▣ A **bar** = one row (open, high, low, close, volume) summarising a sampling interval
- ▣ Time bars: every fixed interval (1 minute, 1 day); oversample quiet periods; fat-tailed returns
- ▣ Tick bars: one bar every N transactions

Volume, dollar and information bars

- ▣ One bar every V shares or every $\$D$ traded: sampling by **information arrival**
- ▣ Returns closer to i.i.d. draws from the Normal distribution: [Clark \(1973\)](#); [Ané & Geman \(2000\)](#)
- ▣ Imbalance / runs bars: sample when order flow becomes one-sided

- ▣ This chapter uses daily time bars; tick data (crypto exchanges) in Chapters 10 and 16

Stationarity and the ADF Test

Definition 10 (Weak stationarity)

- $\{X_t\}$ is (weakly) stationary if its mean, variance and autocovariances do not change over time

Definition 11 (Augmented Dickey–Fuller (ADF) test)

- Regression: $\Delta X_t = \alpha + \rho X_{t-1} + \sum_{j=1}^k \phi_j \Delta X_{t-j} + \varepsilon_t$
 - $H_0: \rho = 0$ (unit root, non-stationary) vs. $H_1: \rho < 0$ (stationary)
 - ADF statistic = t -ratio of $\hat{\rho}$; reject H_0 if it is below the critical value (≈ -2.86 at 5%)
-
- Log prices typically do not reject the unit root; log returns reject it strongly

The Stationarity vs. Memory Dilemma

- ML models need **stationary** inputs: the mapping learned in the past must remain valid
- Prices are non-stationary $\Rightarrow$ practitioners use returns, $\Delta \log P_t$ (integer differentiation, $d = 1$)
- But returns **erase memory**: the level of the price relative to its history is lost

$d = 0$ (prices)

- full memory
- non-stationary

$0 < d < 1$ (fractional)

- stationary
- **memory preserved**

$d = 1$ (returns)

- stationary
- memory erased

- Idea (Hosking (1981); López de Prado (2018)): find the **minimum** d that makes the series stationary

Fractional Differentiation

Definition 12 (Fractional difference operator)

- With the backshift operator $BX_t = X_{t-1}$ and a real order d :

$$(1 - B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-B)^k, \quad \tilde{X}_t = \sum_{k=0}^{\infty} w_k X_{t-k}$$

- Weights obtained recursively:

$$w_0 = 1, \quad w_k = -w_{k-1} \frac{d - k + 1}{k}$$

- $d = 1$: $w = (1, -1, 0, 0, \dots)$ — ordinary returns
- $0 < d < 1$: weights decay slowly (hyperbolically) $\Rightarrow$ long memory is kept

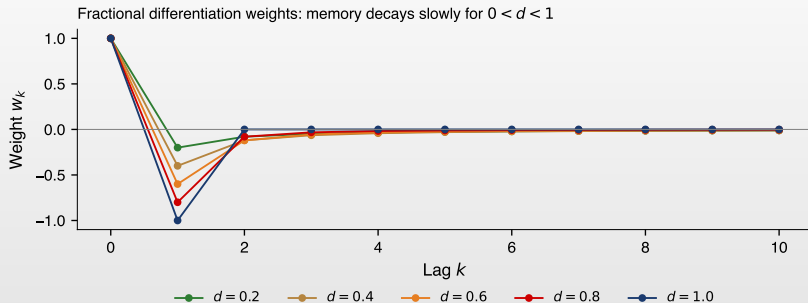
Fixed-Width Window

Definition 13 (Fixed-width-window fractional differentiation (FFD))

- ▣ Truncate the weights: keep $w_0, \dots, w_K$ with $|w_k| \geq \tau$ (e.g. $\tau = 10^{-4}$), drop the rest
- ▣ $\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k}$ for every $t > K$

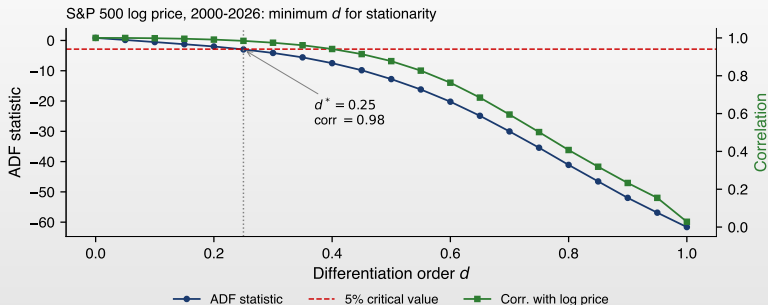
- ▣ Every $\tilde{X}_t$ uses the same number of lags $\Rightarrow$ no drift from a growing window
- ▣ Only past values enter $\Rightarrow$ no look-ahead
- ▣ The smaller d , the more lags are needed before $|w_k| < \tau$
- ▣ The first K observations are lost

Fractional Differentiation Weights



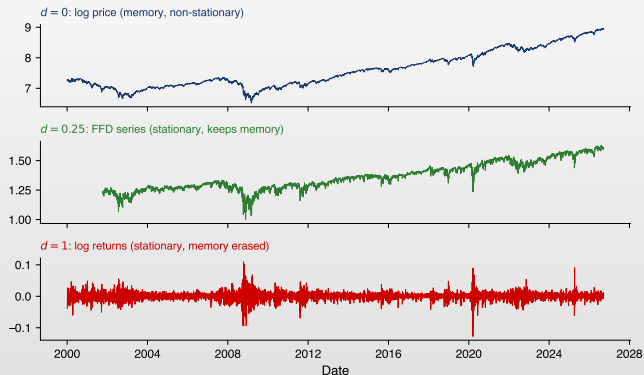
- For $d = 1$ only the first lag matters; for small d the weights of distant lags remain non-zero
- All weights after w_0 are negative: $\tilde{X}_t$ compares today's level with a weighted history

Finding the Minimum d : S&P 500



- ▣ ADF statistic of the FFD log price for $d \in [0, 1]$ (S&P 500, 2000–2026); red line: 5% critical value
- ▣ Green line: Pearson correlation $\text{Corr}(\tilde{X}_t, \log P_t)$ between the FFD series and the original log price
- ▣ Stationarity is reached at $d^* = 0.25$, where the correlation is still 0.98; returns ($d = 1$) keep correlation ≈ 0

Prices, FFD Series and Returns



- **Top:** log price — trending, non-stationary
- **Middle:** FFD with $d^* = 0.25$ — stationary, yet the crashes of 2008 and 2020 remain visible as level shifts
- **Bottom:** log returns — only volatility clusters are visible
- The FFD series enters the case study as feature `ffd_price`

Recap: Financial Data and Features

- ▣ Sampling by activity (volume, dollar bars) gives better-behaved returns than clock time
- ▣ Features must be stationary, but integer differencing throws away memory
- ▣ FFD keeps memory: $d^* = 0.25$ suffices for the S&P 500 (correlation 0.98 with the price)
- ▣ Every transformation must use only past data and be fitted on training folds only
- ▣ Next: how to turn prices into labels

The Fixed-Horizon Label and Its Problems

Definition 14 (Fixed-horizon label)

- h -period log return $r_{t,t+h} = \log P_{t+h} - \log P_t$ and threshold $\tau \geq 0$: $y_t = +1$ if $r_{t,t+h} > \tau$, -1 if $r_{t,t+h} < -\tau$, 0 otherwise
- **Label shares**: fraction of observations in each class, e.g. $\frac{1}{n} \sum_t 1\{y_t = +1\}$

Problems

- A constant τ ignores time-varying volatility: in calm periods almost all labels are 0, in crises almost none
- It ignores the **path**: a real position would have been stopped out long before $t + h$
- Unbalanced classes: S&P 500, $h = 10$ days, $\tau = 0$: 60.5% up vs. 39.5% down

The Triple-Barrier Method

Definition 15 (Triple-barrier label, López de Prado (2018))

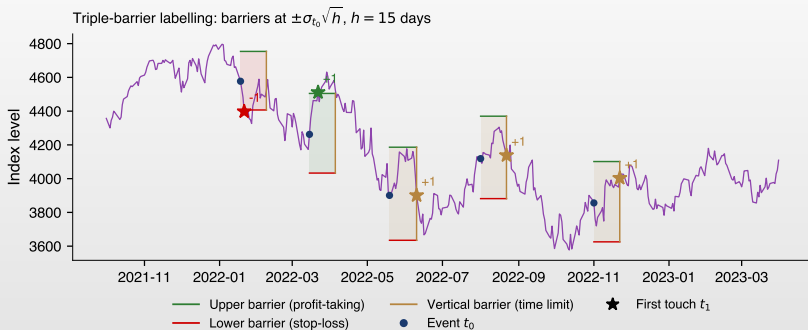
- At event time t_0 , with daily volatility σ_{t_0} and horizon h :
 - ▶ upper barrier (profit-taking): $\log P_t - \log P_{t_0} \geq \pi_u \sigma_{t_0} \sqrt{h}$
 - ▶ lower barrier (stop-loss): $\log P_t - \log P_{t_0} \leq -\pi_l \sigma_{t_0} \sqrt{h}$
 - ▶ vertical barrier (time limit): $t = t_0 + h$
- Label = **first barrier touched** at time t_1 : +1 upper, -1 lower; at the vertical barrier either $\text{sign}(r_{t_0, t_1})$ or 0
- π_u, π_l : barrier widths in units of $\sigma_{t_0} \sqrt{h}$

Volatility Estimate and Properties of the Barriers

Definition 16 (EWMA volatility)

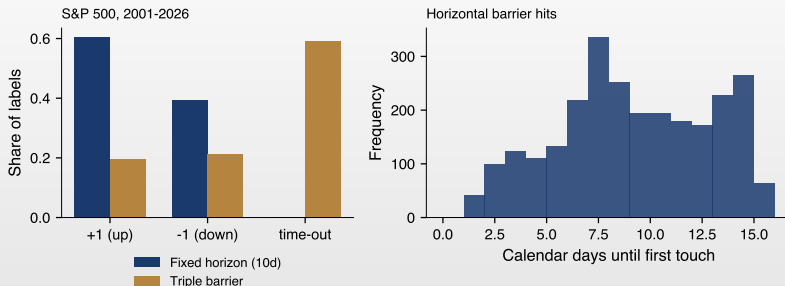
- Exponentially weighted moving average (EWMA) of squared deviations of daily log returns, with span 50 (decay $\lambda = 1 - 2/51$): recent days weigh more
- Barriers scale with volatility $\Rightarrow$ labels are comparable across calm and turbulent regimes
- Mimics how a trader manages a position: take-profit, stop-loss, maximum holding period
- Each label has a **span** $[t_0, t_1]$ — essential for cross-validation (Section 6)

Triple Barrier: S&P 500 in 2022



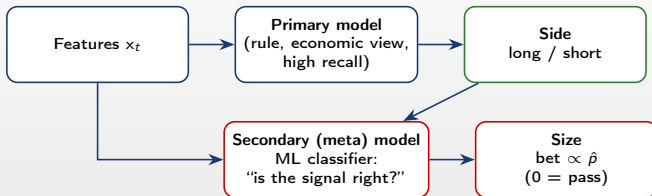
- Five events during the 2022 bear market, $\pi_u = \pi_l = 1$, $h = 15$ days
- January: stop-loss hit (-1); March: profit-taking ($+1$); the others end at the vertical barrier

Label Distribution: Fixed Horizon vs. Triple Barrier



- S&P 500, 2001–2026, $h = 10$: fixed horizon gives 60.5% up / 39.5% down
- Triple barrier ($\pm 1\sigma\sqrt{h}$): 19.6% up, 21.3% down, 59.0% time-out — balanced directional labels
- Right panel: horizontal barriers are typically hit after 1–2 weeks

Meta-Labeling



- Meta-label: $y_t^{meta} = 1$ if the primary signal was profitable (barrier touched in its direction), 0 otherwise
- The secondary model outputs $\hat{p} = P(y_t^{meta} = 1 \mid x_t)$

Why Meta-Labelling Works

- ▣ ML learns **when to trust** the primary model, not the direction itself
- ▣ The primary model is tuned for high recall (catch most opportunities)
- ▣ The meta model filters out false positives $\Rightarrow$ higher precision, and a better F1 score when recall is not sacrificed too much
- ▣ Keeps an interpretable, economically motivated core; limits overfitting of the direction
- ▣ The seminar applies it to a moving-average crossover on Bitcoin

Overlapping Labels and Sample Uniqueness

- Label i spans $[t_{0,i}, t_{1,i}]$; consecutive labels share returns $\Rightarrow$ samples are **not i.i.d.**
- Concurrent labels at t : $c_t = \sum_i 1\{t_{0,i} \leq t \leq t_{1,i}\}$; average uniqueness $\bar{u}_i = \frac{1}{t_{1,i} - t_{0,i} + 1} \sum_{t=t_{0,i}}^{t_{1,i}} \frac{1}{c_t}$
- $\bar{u}_i = 1$: no shared return; $\bar{u}_i = 1/5$: on average five labels overlap

Consequences

- Effective sample size $\approx \sum_i \bar{u}_i \ll n$
- Bagging draws near-duplicates; CV scores too optimistic

Remedies

- Weight samples by $\bar{u}_i$ (and by $|r|$)
- Sequential bootstrap; in RF `max_samples` $\approx \bar{u}$

Recap: Labelling

- ▣ Fixed-horizon labels ignore volatility and the price path, and are unbalanced (60.5% up)
- ▣ Triple-barrier labels scale with volatility and mimic real trade management
- ▣ Meta-labelling lets ML decide *whether* to act on an economic signal
- ▣ Every label has a span; overlapping spans shrink the effective sample
- ▣ Next: validation that respects these spans

Information Leakage: Why Standard K-Fold Fails

Definition 17 (Leakage)

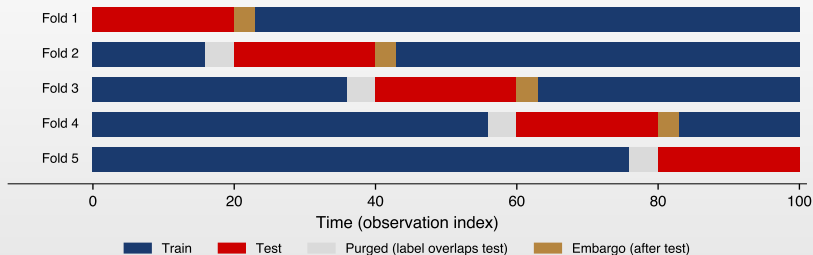
- The training set contains information that is also present in the test set, so the test score no longer measures performance on genuinely new data
- Serially correlated features ($x_t \approx x_{t+1}$) and overlapping labels ($y_t \approx y_{t+1}$): a shuffled K -fold puts **near-identical** days on both sides
- A flexible model (RF, GBM) memorises the neighbour $\Rightarrow$ excellent “out-of-sample” score; tuning on a leaky CV selects the **most overfitted** model
- Other sources: transformations fitted on the full sample (scaling, principal component analysis, the choice of d^*), survivorship bias, data revisions, wrong timestamps

Purging and Embargo

Definition 18 (Purged K-fold with embargo, López de Prado (2018))

- ▣ Folds are **contiguous** blocks in time (no shuffling)
 - ▣ **Purging**: remove from training every observation whose label span $[t_{0,i}, t_{1,i}]$ overlaps the test period
 - ▣ **Embargo**: also remove a small fraction (e.g. 1%) of observations right after the test block, because features look back in time
-
- ▣ Purging handles label overlap; the embargo handles feature memory

Purged K-Fold: The Scheme



- Red: test block; grey: purged observations just before it; amber: embargo just after it
- The training set shrinks a little — the price of an honest estimate

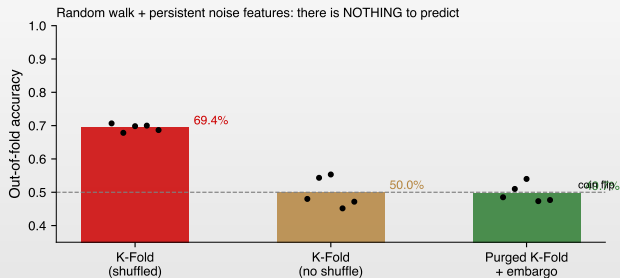
Worked Example: Purging with 10 Observations

- Observations $t = 0, \dots, 9$; label of t ends at $t + 2$ (capped at 9)
- 5 contiguous folds of 2; embargo = 1 observation

Test	Removed	Train
0, 1	2, 3	4–9
2, 3	0, 1, 4, 5	6–9
4, 5	2, 3, 6, 7	0, 1, 8, 9
6, 7	4, 5, 8, 9	0–3
8, 9	6, 7	0–5

- Test $\{4, 5\}$: labels of 2 and 3 end at 4 and 5 $\Rightarrow$ purged; 6 is the embargo; 7 starts when the last test label ends $\Rightarrow$ purged
- With overlapping labels, up to 4 of 10 observations leave the training set

Leakage Experiment: Nothing to Predict



- Price = pure random walk; label = sign of the 20-day forward return
- 5 features = rolling means of independent noise
- True accuracy is 50% by construction
- Random forest classifier, 5 folds

 MFM_ch13_purged_cv

Leakage Experiment: Results

- ▣ **Shuffled K-fold: 69.4%** accuracy — pure leakage
- ▣ Unshuffled (contiguous) K-fold: 50.0%
- ▣ Purged K-fold + embargo: 49.7%

Lesson

- ▣ A 69% hit rate on data with zero signal
- ▣ Never shuffle time series in cross-validation
- ▣ A result that looks too good is usually leakage

Walk-Forward and Combinatorial Purged CV

Walk-forward (WF)

- Train on $[0, T_k]$, test on $(T_k, T_{k+1}]$, roll forward
- Mimics real-time deployment; no future data in training
- But a **single path**: high variance, easy to overfit by repeated trials

Combinatorial purged CV (CPCV)

- N groups; test on all $\binom{N}{k}$ combinations of k groups, with purging and embargo
- $\varphi = \frac{k}{N} \binom{N}{k}$ backtest paths; e.g. $N = 6$, $k = 2$: 15 splits, 5 paths
- A **distribution** of Sharpe ratios instead of one number

- Rule of thumb: purged K-fold / CPCV for model selection; WF for the final, realistic evaluation

Recap: Cross-Validation in Finance

- ▣ Leakage comes from overlapping labels and serially correlated features
- ▣ Shuffled K-fold reports 69.4% on pure noise; purged K-fold reports 49.7%
- ▣ Purging removes overlapping labels; the embargo removes the feature memory after the test block
- ▣ Walk-forward = realistic single path; CPCV = many paths, a distribution of results
- ▣ Next: all of this applied to the S&P 500

Case Study: Data, Target and Validation

- ▣ S&P 500 index and VIX, daily, 2001–2026: 6,269 observations
- ▣ Target: $y_t = 1\{\log P_{t+5} - \log P_t > 0\}$ — the 5-day direction
- ▣ Label span $[t, t + 5]$ used for purging
- ▣ Purged 5-fold CV with 1% embargo
- ▣ Metrics: accuracy and ROC AUC, compared with the “always up” baseline

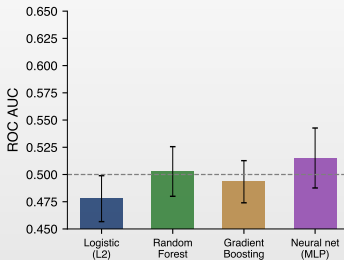
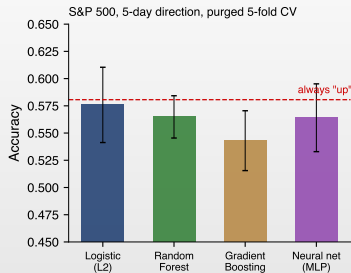
Case Study: 14 Features

- ▣ **Momentum:** $\log P_t - \log P_{t-h}$, $h \in \{1, 5, 20, 60, 120\}$ days
- ▣ **Realised volatility:** annualised standard deviation of daily log returns over 20 and 60 days, and their ratio
- ▣ **Distance to moving average:** $\log P_t$ minus the mean of $\log P$ over the last 50 or 200 days
- ▣ **RSI(14):** $100 - 100/(1 + RS)$, RS = ratio of average gains to average losses over 14 days (smoothed)
- ▣ **VIX:** log level and 5-day change of the implied-volatility index
- ▣ **FFD** log price with $d^* = 0.25$
- ▣ All features use information up to day t only

Case Study: Four Models

- ▣ **Logit**: L2-penalised (Ridge) logistic regression on standardised features
- ▣ **RF**: 400 trees, minimum leaf size 100
- ▣ **GBM**: histogram-based, tree depth 3, learning rate 0.03, 200 trees
- ▣ **MLP**: two hidden layers (32 and 16 units), early stopping
- ▣ Settings deliberately conservative: large leaves and shallow trees limit variance

Model Comparison



	Logit	RF	GBM	MLP
Accuracy	57.6%	56.5%	54.3%	56.4%
AUC	0.478	0.503	0.493	0.515

□ Baseline “always up”: 58.1%

□ No model beats it

Interpreting the Results

Honest conclusion

- ▣ **No model beats the naive “always up” baseline**; all AUCs are close to 0.5
- ▣ Accuracy is misleading with unbalanced classes: 57.6% looks good but is *below* 58.1%
- ▣ Consistent with (weak-form) efficiency of a highly liquid index (Chapter 2)

Where to look instead

- ▣ Cross-sections of many assets (Gu, Kelly & Xiu (2020)); volatility and risk targets
- ▣ Alternative data: text (Chapter 15); less efficient markets: crypto (Chapter 16)
- ▣ Meta-labelling of economically motivated signals

Feature Importance: MDI and MDA

Mean decrease impurity (MDI)

- Average reduction of Gini impurity from all splits on a feature, over the trees of a forest
- In-sample, trees only
- Biased towards continuous, high-cardinality features

Mean decrease accuracy (MDA)

- Permutation importance: drop in an OOS score (here AUC) when the values of a feature are randomly shuffled
- Any model; out of sample
- Can be negative: the feature hurts

- MDI answers “what did the model use?”; MDA answers “what helps on new data?”

Feature Importance: SHAP

Definition 19 (Shapley value of feature j (SHAP, Lundberg & Lee (2017)))

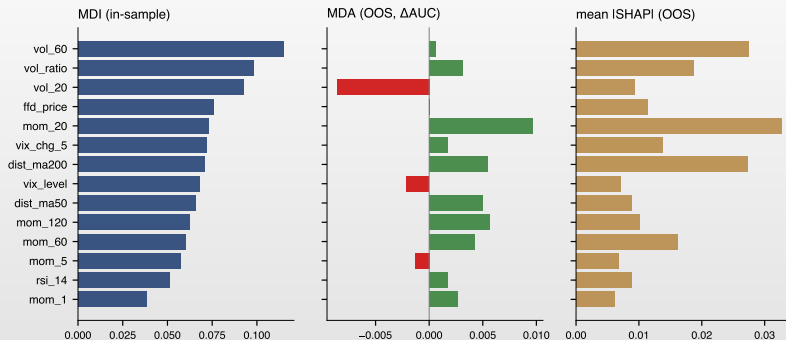
- Average marginal contribution of j over all coalitions $S \subseteq \{1, \dots, p\} \setminus \{j\}$:

$$\phi_j = \sum_S \frac{|S|! (p - |S| - 1)!}{p!} [v(S \cup \{j\}) - v(S)], \quad \hat{f}(x) = \phi_0 + \sum_j \phi_j$$

- $v(S)$: model prediction using only the features in S
- Additive attribution of each prediction; mean $|\phi_j|$ over observations = global importance
- **Substitution effects**: correlated features share (and dilute) importance — cluster them first

 Shapley Values

Feature Importance in the Case Study



- Three panels: MDI (in-sample), MDA (ΔAUC out of sample), mean |SHAP| (out of sample)
- Same random forest and features as in the model comparison

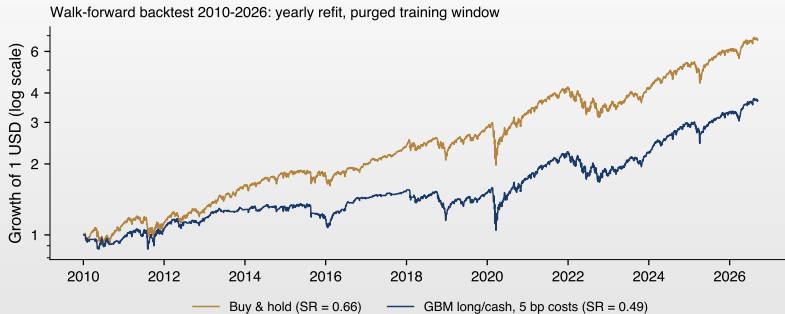
Feature Importance: Interpretation

- ▣ **MDI** ranks volatility features first: vol_60, vol_ratio, vol_20
- ▣ **MDA** and **SHAP** point to mom_20 and dist_ma200
- ▣ vol_20: high MDI but **negative** MDA — the model uses it to fit noise
- ▣ Importances are tiny ($\Delta\text{AUC} < 0.01$), consistent with the model comparison
- ▣ **Backtesting is not a research tool**, importance analysis is: [López de Prado \(2018\)](#)

Walk-Forward Backtest: Design

- Model: GBM refitted every January on all past data, with purging of labels that overlap the test year
- Rule: long the S&P 500 if $\hat{p} > 0.5$, otherwise cash ($w_t \in \{0, 1\}$)
- Costs: 5 bp per unit of turnover
- Out-of-sample period: 2010–2026, one refit per year
- Benchmark: buy-and-hold ($w_t = 1$)

Walk-Forward Backtest: Results



- 2010–2026: strategy Sharpe **0.49** vs. buy-and-hold **0.66** (annualised); average exposure 77%
- The model mostly learned “be long”, and its exits came at the wrong times

Recap: Case Study

- ▣ Four models, 14 features, 25 years, leakage-free validation
- ▣ No model beats “always up” (58.1%); AUCs between 0.478 and 0.515
- ▣ Feature importances are tiny and differ between in-sample and out-of-sample measures
- ▣ The walk-forward strategy underperforms buy-and-hold after costs (Sharpe 0.49 vs. 0.66)
- ▣ An honest negative result is a result

Multiple Testing and Selection Bias

- ▣ A **backtest**: simulation of a strategy on historical data
- ▣ A researcher tries N strategies or configurations and reports the best one
- ▣ Even if all have zero true Sharpe ratio, $\max_n \widehat{SR}_n$ grows with N
- ▣ **Selection bias**: the reported maximum overstates the true performance
- ▣ [Harvey, Liu & Zhu \(2016\)](#): given hundreds of published factors, a new factor needs a t -statistic above 3, not 2

The False Strategy Theorem

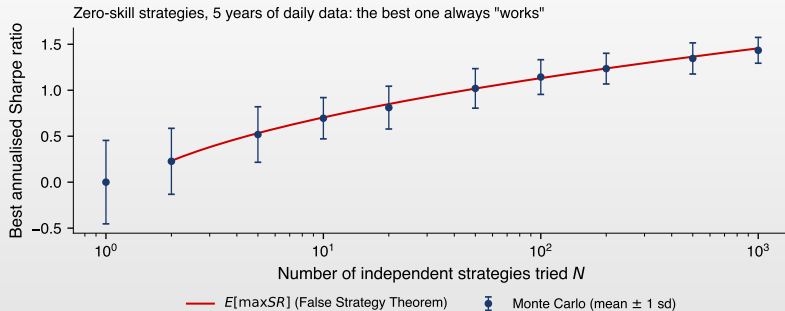
Theorem 1 (False Strategy Theorem, Bailey & López de Prado (2014))

- Let $\widehat{SR}_n$, $n = 1, \dots, N$, be independent estimates with $\mathbb{E}[\widehat{SR}_n] = 0$ and variance $\text{Var}[\widehat{SR}]$; then the **expected maximum Sharpe ratio** is

$$\mathbb{E}\left[\max_n \widehat{SR}_n\right] \approx \sqrt{\text{Var}[\widehat{SR}]} \left[(1 - \gamma) \Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma \Phi^{-1}\left(1 - \frac{1}{N_e}\right) \right]$$

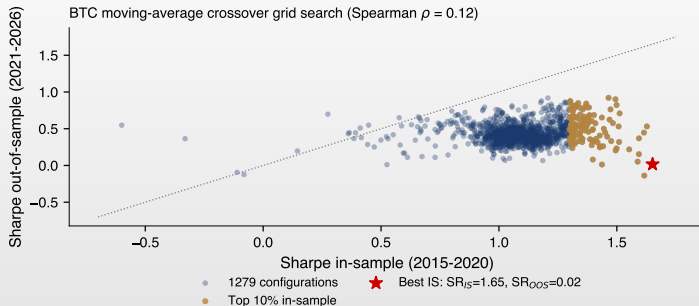
- $\gamma \approx 0.5772$: Euler–Mascheroni constant; Φ : standard Normal cumulative distribution function
- For i.i.d. returns, $\text{Var}[\widehat{SR}] \approx 1/T$ per period

The Best of N Zero-Skill Strategies



- Strategies with i.i.d. zero-mean returns, 5 years of daily data (annualised $\text{Var}[\widehat{SR}] \approx 1/5$)
- Trying $N = 1000$ variants yields a best Sharpe ratio of ≈ 1.45 — **with no skill at all**
- The Monte Carlo simulation agrees with the theorem

In-Sample vs. Out-of-Sample: Bitcoin Grid Search



- Moving-average (MA) crossover on Bitcoin: long when the fast MA of the log price exceeds the slow MA by more than a band
- 1,279 configurations (fast window, slow window, band); in-sample 2015–2020, out-of-sample 2021–2026

Bitcoin Grid Search: Interpretation

- ▣ Best IS configuration (14/20 days): $SR_{IS} = 1.65 \rightarrow SR_{OOS} = 0.02$ (annualised with $\sqrt{365}$)
- ▣ Spearman rank correlation between IS and OOS Sharpe ratios: $\rho = 0.12$
- ▣ Median OOS SR: 0.53 for the top-10% IS configurations vs. 0.42 for all
- ▣ The *family* of trend rules carries some information, but the *selected* optimum is noise

Probabilistic Sharpe Ratio

Definition 20 (Probabilistic Sharpe Ratio (PSR), Bailey & López de Prado (2012))

- Observed per-period $\widehat{SR}$ from T returns, skewness $\hat{\gamma}_3$, kurtosis $\hat{\gamma}_4$ (Normal distribution: $\gamma_4 = 3$):

$$\widehat{PSR}(SR^*) = \Phi \left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right)$$

- Skewness $\gamma_3 = \mathbb{E}[(r - \mu)^3]/\sigma^3$; kurtosis $\gamma_4 = \mathbb{E}[(r - \mu)^4]/\sigma^4$ (Chapter 1)
- PSR = probability that the true SR exceeds a benchmark SR^* , given sample length and non-Normality

Deflated Sharpe Ratio

Definition 21 (Deflated Sharpe Ratio (DSR), Bailey & López de Prado (2014))

□ $DSR = \widehat{PSR}(SR_0)$, with the benchmark set to the expected maximum under the null of no skill:

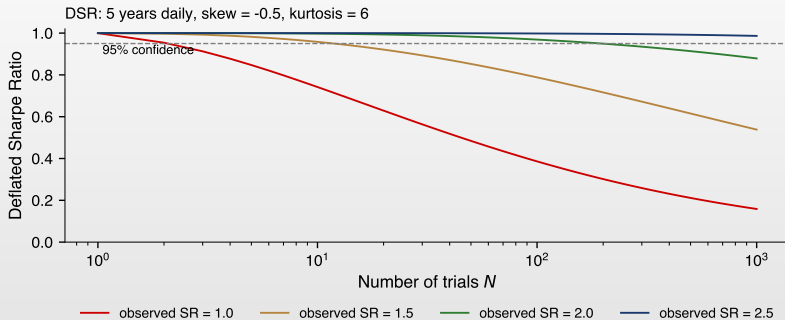
$$SR_0 = \sqrt{\text{Var}[\widehat{SR}_n]} \left[(1 - \gamma) \Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma \Phi^{-1}\left(1 - \frac{1}{N_e}\right) \right]$$

- DSR corrects for **selection bias** (N trials), **non-Normality** and **sample length**
- Reject the strategy unless $DSR > 0.95$; report N honestly

Worked Example: DSR of the Best Bitcoin Rule

- ▣ Best IS rule: $T = 2,297$ daily returns, $\widehat{SR} = 0.0864$ per day (1.65 annualised), $\hat{\gamma}_3 = 0.10$, $\hat{\gamma}_4 = 12.71$
- ▣ Denominator: $\sqrt{1 - 0.10 \cdot 0.0864 + \frac{12.71-1}{4} \cdot 0.0864^2} = 1.007$
- ▣ $N = 1,279$ trials. Two choices of $\text{Var}[\widehat{SR}_n]$:
 - ▶ dispersion of the 1,279 trial SRs (std. 0.0097): $SR_0 = 0.032/\text{day}$, $z = 2.58$, $DSR = 0.995$
 - ▶ independent-trials value $1/T$ (std. 0.0209): $SR_0 = 0.069/\text{day}$, $z = 0.81$, $DSR = 0.792 < 0.95$
- ▣ The trials are highly correlated (neighbouring windows), so their dispersion understates the true search: the verdict depends on this choice
- ▣ The out-of-sample Sharpe of 0.02 sides with the conservative value

Deflated Sharpe Ratio in Practice



- 5 years of daily returns, skewness -0.5 , kurtosis 6
- An annualised SR of 1.0 is significant only if (almost) no other strategy was tried: it fails after 2–3 trials
- $SR = 1.5$ survives ≈ 10 trials, $SR = 2.0 \approx 200$; only $SR = 2.5$ survives 1000

Probability of Backtest Overfitting (PBO)

Combinatorially symmetric cross-validation (CSCV): Bailey, Borwein, López de Prado & Zhu (2017)

1. Matrix M of returns: T periods $\times$ N strategy configurations
2. Split rows into S blocks; for each of the $\binom{S}{S/2}$ combinations, form IS (half the blocks) and OOS (the other half)
3. Select the best configuration n^* in-sample; compute its relative rank $\bar{\omega} \in (0, 1)$ out-of-sample
4. Logit $\lambda = \log \frac{\bar{\omega}}{1-\bar{\omega}}$
5. $PBO = \Pr[\lambda < 0]$: probability that the IS winner ranks below the OOS median

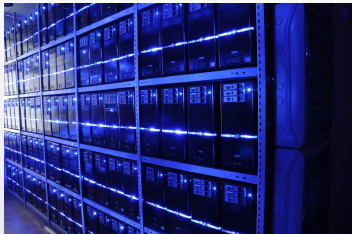
Reading the PBO

- ▣ $PBO \approx 0$: the in-sample winner reliably stays above the median out of sample
- ▣ $PBO \approx 0.5$ or higher: the selection process has no skill — choosing the IS winner is like choosing at random
- ▣ PBO complements DSR: it evaluates the **research procedure**, not a single strategy
- ▣ Both require logging **every** trial, including the failed ones

Recap: Backtest Overfitting

- ▣ The best of many backtests is biased upwards: 1000 zero-skill trials give a Sharpe of ≈ 1.45
- ▣ Bitcoin grid: IS winner 1.65 $\rightarrow$ OOS 0.02
- ▣ PSR corrects for sample length and non-Normality; DSR also for the number of trials
- ▣ PBO measures the reliability of the whole selection procedure
- ▣ Record N ; deflate before you claim

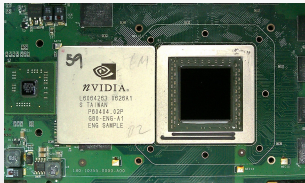
The Hardware Behind Modern ML



A data centre

Photo: BalticServers.com (2013); CC BY-SA

3.0; Wikimedia Commons



NVIDIA G80 graphics processor

Photo: Hyins (2011); public domain;

Wikimedia Commons

- Deep networks and time-series foundation models (Chapter 14) are trained on **GPUs** (graphics processors) in large data centres
- A GPU runs thousands of simple cores in parallel: ideal for matrix products
- In finance the binding constraint is usually the data (one history, low signal-to-noise), not the compute

Deep Learning for Sequences

Recurrent networks (LSTM, GRU)

- ▣ Hidden state carries memory: $h_t = g(h_{t-1}, x_t)$
- ▣ Gates control what is kept or forgotten, mitigating vanishing gradients
- ▣ Used for volatility, limit-order-book and intraday forecasting

Transformers

- ▣ Self-attention: $\text{softmax}(QK^\top / \sqrt{d_k})V$, with query, key, value matrices Q, K, V
- ▣ Long-range dependence, parallel training
- ▣ Backbone of large language models (Chapter 15)

Time-Series Foundation Models

Pre-trained forecasters

- ▣ Pre-trained on large collections of series, used **zero-shot** (without re-training): Chronos (Ansari et al., 2024), TimesFM (Das et al., 2023), Moirai (Woo et al., 2024), Lag-Llama (Rasul et al., 2023)
 - ▣ Useful baselines for volatility and volume forecasting
 - ▣ For returns they inherit the same signal-to-noise limits
 - ▣ Pre-training data may overlap your test period: a new form of leakage
-
- ▣ Covered in depth, with the risk-forecast evaluation tools of Chapter 8, in Chapter 14

Reinforcement Learning, Generative Models, XAI, Risk

Reinforcement learning (RL)

- A policy $\pi(a | s)$ chooses actions a in states s to maximise cumulative reward
- Optimal execution, market making, deep hedging: Buehler et al. (2019)

Generative models

- GANs, variational autoencoders (VAEs), diffusion models: synthetic scenarios for stress tests

Explainable AI (XAI)

- Required by regulators (EU AI Act, credit scoring, model risk)
- SHAP, LIME (local interpretable model-agnostic explanations), partial dependence

ML for risk

- Volatility, Value-at-Risk, Expected Shortfall with quantile forests or quantile-loss boosting (Chapters 7–8)

- Quantinar:  Reinforcement Learning  GANs  Explainable AI  ML in Financial Risk

Best-Practice Checklist

Data and features

- ☐ Point-in-time data, no survivorship bias
- ☐ Stationary features that keep memory: FFD
- ☐ Transformations fitted on training folds only

Labels

- ☐ Volatility-scaled, path-dependent labels
- ☐ Record label spans; weight by uniqueness

Validation

- ☐ Purged K-fold / CPCV with embargo
- ☐ Always compare with naive baselines
- ☐ Report AUC and log-loss, not only accuracy

Backtesting

- ☐ Include transaction costs and slippage
- ☐ Log every trial; report N
- ☐ Deflated Sharpe ratio and PBO before any claim

Conclusions

- ▣ ML offers flexible models, but financial data offer **little signal and much noise**
- ▣ Classical tools (LASSO, random forests, boosting, neural nets) remain the workhorses; the finance-specific part is the **process**
- ▣ Fractional differentiation reconciles stationarity and memory ($d^* = 0.25$ for the S&P 500)
- ▣ Triple-barrier and meta-labelling produce labels that reflect real trading
- ▣ Shuffled K-fold can report 69% accuracy on pure noise: use purging and embargo
- ▣ On the S&P 500, no model beats “always up”; the best of many backtests must be deflated

Next

- ▣ Chapter 14: deep learning and time-series foundation models; Chapter 15: large language models and sentiment

Further Study on Quantinar

Advanced courses on quantinar.com

-  Random Forests
-  Shapley Values
-  Shapley Curves: A Smoothing Perspective
-  Machine Learning in Financial Risk
-  Introduction to Reinforcement Learning
-  Generative Adversarial Networks
-  MSCA Digital: The Need for Explainable AI
-  Business Analytics and Data Science
-  Model Selection Criteria
-  PAC Learning

References (1/3)

- ▣ Ané, T., Geman, H. (2000). Order flow, transaction clock, and normality of asset returns. *Journal of Finance*, 55(5), 2259–2284.
- ▣ Ansari, A.F. et al. (2024). Chronos: learning the language of time series. *arXiv:2403.07815*.
- ▣ Bailey, D.H., Borwein, J., López de Prado, M., Zhu, Q.J. (2017). The probability of backtest overfitting. *Journal of Computational Finance*, 20(4), 39–69.
- ▣ Bailey, D.H., López de Prado, M. (2012). The Sharpe ratio efficient frontier. *Journal of Risk*, 15(2), 3–44.
- ▣ Bailey, D.H., López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- ▣ Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- ▣ Buehler, H., Gonon, L., Teichmann, J., Wood, B. (2019). Deep hedging. *Quantitative Finance*, 19(8), 1271–1291.
- ▣ Clark, P.K. (1973). A subordinated stochastic process model with finite variance for speculative prices. *Econometrica*, 41(1), 135–155.
- ▣ Das, A., Kong, W., Sen, R., Zhou, Y. (2023). A decoder-only foundation model for time-series forecasting. *arXiv:2310.10688*.

References (2/3)

- Dixon, M.F., Halperin, I., Bilokon, P. (2020). *Machine Learning in Finance: From Theory to Practice*. Springer.
- Friedman, J.H. (2001). Greedy function approximation: a gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.
- Gu, S., Kelly, B., Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.
- Harvey, C.R., Liu, Y., Zhu, H. (2016). ...and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Hastie, T., Tibshirani, R., Friedman, J. (2009). *The Elements of Statistical Learning*. 2nd ed. Springer.
- Hoerl, A.E., Kennard, R.W. (1970). Ridge regression: biased estimation for nonorthogonal problems. *Technometrics*, 12(1), 55–67.
- Hosking, J.R.M. (1981). Fractional differencing. *Biometrika*, 68(1), 165–176.
- Kelly, B., Xiu, D. (2023). Financial machine learning. *Foundations and Trends in Finance*, 13(3–4), 205–363.

References (3/3)

- ▣ Lo, A.W. (2004). The adaptive markets hypothesis. *Journal of Portfolio Management*, 30(5), 15–29.
- ▣ López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
- ▣ López de Prado, M. (2020). *Machine Learning for Asset Managers*. Cambridge University Press.
- ▣ Lundberg, S.M., Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.
- ▣ Rasul, K. et al. (2023). Lag-Llama: towards foundation models for probabilistic time series forecasting. *arXiv:2310.08278*.
- ▣ Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society B*, 58(1), 267–288.
- ▣ Woo, G. et al. (2024). Unified training of universal time series forecasting transformers. *arXiv:2402.02592*.
- ▣ Zou, H., Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society B*, 67(2), 301–320.
- ▣ **Code and data:** github.com/danpele/MFM/tree/main/Quantlets/Ch_13
- ▣ Campbell, M., Hoane, A.J., Hsu, F. (2002). Deep Blue. *Artificial Intelligence*, 134(1–2), 57–83.
- ▣ Rosenblatt, F. (1958). The perceptron: a probabilistic model for information storage and organization in the brain. *Psychological Review*, 65(6), 386–408.